

Vine Vanguard

The Egg Crusher: A Scalable Solution for Spotted Lanternfly Egg Management

Jackie Antelo, Josie Chong, Christopher Guillen-Chacon, Alix Martini, Ryan Sigler

Problem Statement & Context:

Vineyard owners across New York State are affected by the rapid spread of Spotted Lanternflies (SLF), whose presence in the Lake Erie and Finger Lakes regions alone could have projected losses reaching approximately \$8.8 million within three years [1]. A single SLF egg mass contains 30–50 eggs, and populations can spread rapidly across large areas of vineyard land. To effectively address this spread, reduce proliferation, and prevent these economic losses from spreading to other parts of the nation, these invasive insects should be targeted while in their egg stage. Destroying eggs prevents the flies from maturing into reproductive adults, reducing the number of SLF that will lay future egg masses and limiting their exponential population growth. Our team focused on developing a mechanical solution for large-scale egg mass destruction that can be deployed at the vineyard level.

Current elimination methods are not suitable for addressing large-scale SLF populations. Manually crushing individual egg masses is time-consuming and inconsistent. Submersion into rubbing alcohol requires constant refilling and storage capacity. Experimental oil ovicides are only effective on 75% of eggs [2]. Additional consideration must be given to the consequences of ineffective elimination, as even one surviving egg mass can enable SLF persistence, escalating plant damage and contaminated harvests.

There is a need for a systematic method for eliminating large quantities of SLF egg masses that can be integrated into existing collection workflows. This method must be consistently effective, practical for real-world use, while minimizing storage, disposal, and labor demands on vineyard operations during implementation.

Final Prototype and Application:

We developed a device that mechanically crushes eggs, utilizing a juicer-inspired crushing mechanism.

How it works:

- Eggs masses enter the mesh cylinder chamber through an inlet funnel.
- A rotating spiral, driven by an electric motor, transports the egg masses downward while simultaneously pressing the eggs against the mesh cylinder, where the crushing occurs.
- A sweeper attached to the rotating spiral pushes crushed eggs towards a hole in the bottom of the mesh housing, falling out of the device through an outlet funnel.

This device will be deployed in the field alongside the given collection method that best suits the vineyard's needs, such as a manual scraper or an autonomous scraping robot. The device is modular, such that its method of accepting and ejecting egg masses can be easily changed.

Conclusion and Recommendation:

We would recommend the continued research and development of the Egg Crusher for applications in vineyards. The device in its current form is a successful proof of concept for a scalable, efficient, and practical egg extermination method, demonstrating basic mechanical motion. Its ~80% crushing efficiency is on par with current methods, without the drawbacks of consistent maintenance or manual labor. Additionally, its 100% efficiency for ejecting crushed eggs of a standardized diameter

suggests the crushed eggs can be re-incorporated into the ecosystem as a compost ingredient if desired by the vineyard owners.

We would suggest three primary redesigns in future phases of this project:

1. *Spiral and sweeper geometry.* Pitch angle, spiral diameter, and other features can be optimized to improve crushing efficiency and properly push the crushed eggs towards the outlet funnel.
2. *Compactness.* The system can be scaled down to reduce weight and improve portability, making it easier to deploy in the field alongside a collection device.
3. *Contingency mechanisms.* Incorporating multiple built-in crushing mechanisms arranged in a linear series can improve crushing efficiency, as eggs not crushed by the first spiral will be crushed by the second crushing mechanism right below.

Testing and Results:

Using hydrated Orbeez as model SLF eggs due to their comparable size and material properties as well as their uniformity, we were able to test:

1. *The dependence of the quantity of inputted eggs on machine efficiency.*

We wanted to determine if the system had an optimal loading range. Results showed that efficiency decreases as quantity increases beyond the optimal operating load. Trials with 50 eggs produced a crushing efficiency range of 72-92% with an average crushing efficiency of 82%. With trials of 100 eggs, the crushing efficiency dropped to a range of 64-78% with an average crushing efficiency of 72%. This indicates that overcrowding inside the chamber restricts motion and reduces effective processing, limiting throughput at higher loads. However, it does also indicate that the device is suitable for crushing one egg mass at a time which contains anywhere from 30-50 eggs.

2. *The dependence of operation time on machine efficiency.*

We wanted to determine whether longer runtime improves performance. Across both 90 second and 3 minute trials with 100 eggs, results showed minimal or no improvement over time. Crushing efficiency remained within similar ranges of 64-78% with an average crushing efficiency of 72% for 90 seconds and 65-78% with an average crushing efficiency of 71% for 3 minutes. This suggests that the system reached a steady-state operating condition quickly which means that extending the operating time of the system does not significantly enhance crushing performance once equilibrium is reached. This indicates that our device can limit the amount of energy used as increasing the operating time does not increase the crushing efficiency.

3. *The system failure rate.*

Finally, we evaluated the system failure rate which we defined as eggs that made it through the device uncrushed. Across all trials, with our standard Orbeez model egg size of $\frac{1}{8}$ ", no uncrushed eggs were observed indicating a 0% failure rate under these standardized conditions. In the cases where an uncrushed Orbeez made it through the device, the Orbeez tended to be less hydrated and smaller measuring closer to $\frac{1}{16}$ " rather than $\frac{1}{8}$ ".

Key outcomes:

- Optimal performance occurs at lower to moderate input quantities as the 50-100 eggs range shows diminishing returns as the load increases
- Time is not a limiting factor after initial steady-state is reached, 90 seconds seems to be sufficient
- System reliability is high, with effectively complete crushing under tested conditions

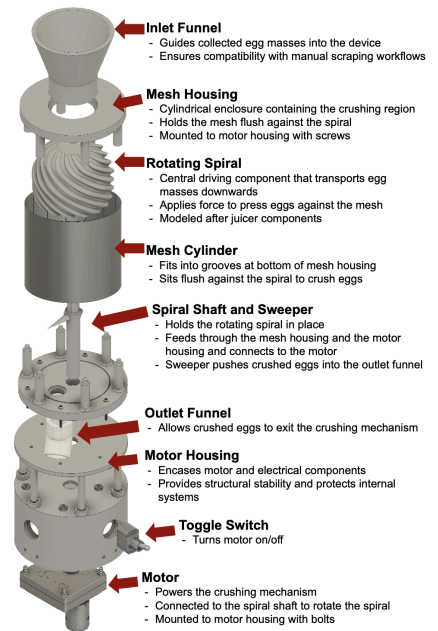
Prototype and Testing Details:

Our design draws inspiration from juicer systems, where material is guided through a rotating spiral and compressed against an abrasive mesh surface. Similarly, our device crushes input SLF egg masses by forcing them downward through a rotating central spiral against a fixed mesh barrier. This approach prioritizes compactness, portability, and compatibility with future system integration.

Following initial mock-ups to determine overall dimensions and structural layout, we developed multiple iterative prototypes to address challenges in spiral alignment, clearance tolerances, and material flow. The final functional prototype is shown in the figure below, which illustrates the full system architecture and component interactions.

Component Specifications and Assembly:

The final prototype consists of nine 3D-printed PLA components fabricated using the Rapid Prototyping Lab, integrated with purchased mechanical and electrical components. The central crushing mechanism utilizes a McMaster 6409K17 compact DC gearmotor (12V DC, 26 rpm, 320 in-oz torque) to drive a custom-designed spiral shaft assembly. The crushing chamber features a cylindrical mesh housing that secures a 304 stainless steel mesh cylinder (20x20 mesh, 0.034" openings) flush against the rotating spiral. For enhanced filtration, a secondary fine mesh layer (270x270 mesh, 0.0023" openings) provides additional crushing precision. The spiral geometry was iteratively optimized through multiple prototypes to achieve proper clearance tolerances and material flow characteristics. Assembly is achieved through M3 x 25mm Phillips head screws (McMaster 99461A948) that secure the mesh housing to the motor housing, while the motor itself is mounted with dedicated bolts. A toggle switch (SPST-NO, 6A rating, McMaster 7343K184) provides simple on/off control. The modular design allows for easy disassembly for maintenance or component replacement. Material flow follows a consistent path: egg masses enter through the inlet funnel, are transported downward by the spiral while being compressed against the mesh barrier, and crushed material exits via the sweeper-directed outlet funnel. The entire assembly weighs approximately 2.5 pounds and measures roughly 12 inches in height, making it suitable for field deployment alongside existing collection workflows.



References

- [1] A. F. Pinto, B. C. Eshenaur, F. E. Acevedo, A. A. Calixto, M. Centinari, and M. I. Gómez, "Assessing the potential economic impacts of spotted lanternfly (Hemiptera: Fulgoridae) infestations on grape production in New York State," *Journal of Integrated Pest Management*, vol. 16, no. 1, 2025. [Online]. Available: <https://doi.org/10.1093/jipm/pmae039>
- [2] G. Krawczyk, "What should you do with spotted lanternfly egg masses?" Penn State Extension. [Online]. Available: <https://extension.psu.edu/what-should-you-do-with-spotted-lanternfly-egg-masses>

Utilized in the Prototype						
Part Name	Quantity	Unit of Measurement	Source	Specs	Supplier/Process	Cost (\$)
Spiral	1	Each	Fabricated	238.500 grams, Light Gray PLA	RPL (3D Print)	11.93
Spiral Shaft	1	Each	Fabricated	22.571 grams, Light Gray PLA	RPL (3D Print)	1.13
Mesh Housing (Top)	1	Each	Fabricated	122.384 grams, Light Gray PLA	RPL (3D Print)	6.12
Mesh Housing (Bottom)	1	Each	Fabricated	143.625 grams, Light Gray PLA	RPL (3D Print)	7.18
Inlet Funnel	1	Each	Fabricated	100.636 grams, Light Gray PLA	RPL (3D Print)	5.03
Motor Housing (Top)	1	Each	Fabricated	116.623 grams, Light Gray PLA	RPL (3D Print)	5.83
Motor Housing (Bottom)	1	Each	Fabricated	325.375 grams, Light Gray PLA	RPL (3D Print)	16.27
Switch Holder	1	Each	Fabricated	6.118 grams, Light Gray PLA	RPL (3D Print)	0.31
Sweeper	1	Each	Fabricated	0.578 grams, Light Gray PLA	RPL (3D Print)	0.03
Motor	1	Each	Purchased	Compact DC Gearmotor, Square Face, 12V DC, 26 rpm @ 320 in.-oz.	McMaster (6409K17)	72.64
Switch	1	Each	Purchased	Toggle Switch, 2 Position, Round, Maintained, 2 Terminal, SPST-NO, 6A, Tab Terminal	McMaster (7343K184)	8.96
Mesh Cylinder	1 x 1	Feet	Purchased	304 Stainless Steel Filter Mesh, 20 x 20 Mesh Size, 0.034" Opening Size, 1 ft x 1 ft	McMaster (85385T73)	15.2
Outlet Funnel	1	Each	Purchased	Standard-Wall PVC Pipe 45 Degree Elbow Connector, White, 3/4 Socket, Connect Female	McMaster (4880K32)	1.45
M3 Screws	1	Pack of 50 Each	Purchased	Phillips Rounded Head Screws for Plastic, 18-8 Stainless Steel, M3 Screw Size, 25 mm Long	McMaster (99461A948)	10.91
Fine Mesh	1 x 2	Feet	Purchased	Stainless Steel, 270 x 270 mesh size, 0.0023" Opening	McMaster (85385T113)	32.57
Total Prototype Price						195.56

Additional Project Expenses						
Part Name	Quantity		Source	Specs	Supplier/Process	Cost (\$)
Mesh Housing (Bottom) - 0	1	Each	Fabricated	116.744 grams, Light Gray PLA	RPL (3D Print)	5.84
Mesh Housing (Top) - 0	1	Each	Fabricated	146.875 grams, Light Gray PLA	RPL (3D Print)	7.34
Spiral - 0	1	Each	Fabricated	198.400 grams, Light Gray PLA	RPL (3D Print)	9.92
Shaft - 0	1	Each	Fabricated	11.988 grams, Light Gray PLA	RPL (3D Print)	0.6
Motor Housing	1	Each	Fabricated	275.303 grams, Light Gray PLA	RPL (3D Print)	13.77
Spiral - 2	1	Each	Fabricated	241.859 grams, Light Gray PLA	RPL (3D Print)	12.09
Total Additonal Price						49.56

Total Project Cost	Cost (\$)
Prototype	195.56
Additional	49.56
Grand Total	245.12

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Problem and Context

Spotted Lanternfly (SLF) infestation is rapidly spreading across vineyards

- Populations grow quickly due to high reproduction rates
- Egg masses are laid directly on grapevines and surrounding surfaces
- Each female lays **30–120 eggs per year**¹
- Heavily infested vines can host **~400 SLF per vine**¹

Current egg extermination methods are insufficient

- Manual crushing is **time-consuming and inconsistent**
- Liquid submersion requires **constant maintenance** and refilling
- Chemical ovicides achieve only **~75% effectiveness**¹

Why it matters:

Vineyard owners require a method that is:

- **Efficient** at a large scale
- Consistently **effective**
- **Practical** for real-world use

Solution Process

Design Approach:

MOCK UP → FUNCTIONAL PROTOTYPE → FINAL PRODUCT

- Iterative prototyping process:

- Inspired by **juicer mechanisms** to crush SLF egg masses
- Focus on a compact and mobile device

Key Features of Functional Prototype:

- Rotating spiral to move and crush eggs
- Mesh barrier for crushing
- Modular structure for future motor integration

Limitations and Challenges:

- Spiral misalignment reduces crushing efficiency
- Egg accumulation beneath the spiral

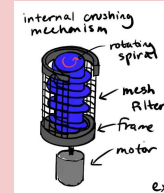


Fig. 1 and 2: Sketch and Mock-Up of Spiral Mechanism. Initial sketches from the mock-up stage of the internal crushing mechanism utilizing the rotating spiral and the cardboard prototype

Testing the Final Prototype

Modeling the Eggs:

To simulate SLF egg masses in a controlled and repeatable way, we used **hydrated Orbeez** as a physical model system

- **Size similarity:** Measurements show approximately 6 Orbeez per inch while real SLF egg masses contain approximately 7-8 eggs per inch

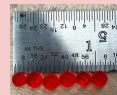


Fig. 3: Image of Orbeez. Approximately 6 hydrated Orbeez makes up 1 inch



Fig. 4: Image of SLF Egg Masses. Each egg mass is about 1 inch long and 7-8 eggs make up each "column"

- **Material properties:** the soft and deformable structure mimics the texture of SLF eggs
- **Consistency and repeatability:** Uniform size and shape allow for repeatable trials

Test 1: How Does Quantity Impact Crusher Efficiency?

- Crushing efficiency typically **decreases as quantity increases beyond an optimal load**
- Higher quantities lead to **overcrowding inside the crusher**, limiting effective egg movement
- Material flow becomes restricted
- Slower movement through the crusher reduces processing efficiency

Key Takeaways:

The crusher is **most effective at lower quantities**, making it well-suited for eliminating individual SLF egg masses, which typically contain about 30–50 eggs¹.

Test 2: How Does Operation Time Impact Crusher Efficiency?

- Crushing efficiency shows **little to no significant improvement as operation time increases**
- After an initial period, the system appears to reach a **steady state** where additional time doesn't meaningfully enhance results
- Most of the effective crushing likely occurs early on. This suggests that time is not a limiting factor in improving efficiency

Key Takeaways:

Crushing efficiency remains **near constant over time**, demonstrating that extending operation time does not meaningfully improve performance once the system reaches its effective operating capacity.

Test 3: How Many Uncrushed Eggs Are Ejected?

- Throughout trials, **no uncrushed eggs** of diameter greater than 1/16" were ejected from the system

Key Takeaways:

With respect to our standardized egg approximation of 1/8" diameter, our crusher **consistently crushes all ejected eggs**.

Final Design

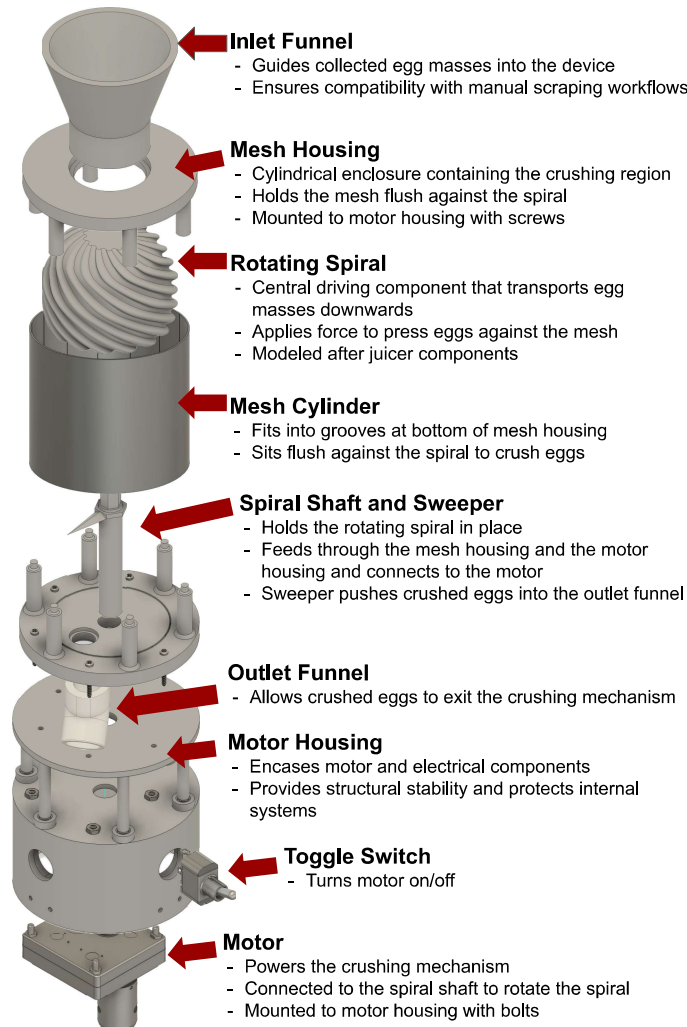


Fig. 8: Exploded Diagram. Each component's CAD file sized proportionally then assembled into a single diagram to show approximate locations, sizes, and material of each component.

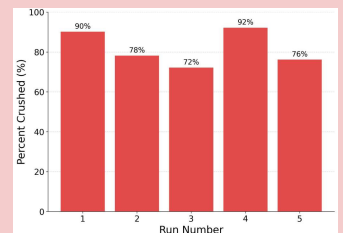


Fig. 5: Plot of Trial 1: 50 SLF Eggs for 90 Seconds. Crushing efficiency ranged from 72% to 92% over five runs, with generally high efficiency and some variability between trials.

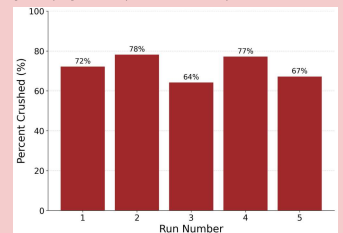


Fig. 6: Plot of Trial 2: 100 SLF Eggs for 90 Seconds. Crushing efficiency ranged from 64% to 78%, showing a slight decrease compared to the lower quantity.

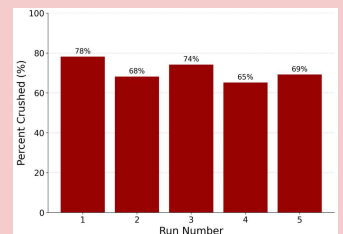


Fig. 7: Plot of Trial 3: 100 SLF Eggs for 3 Minutes. Crushing efficiency ranged from 65% to 78%, showing little to no improvement compared to the 90-second trials.

Conclusions and Recommendations

- The device successfully demonstrates **basic mechanical motion** and **~80% efficiency** for the utilized crushing mechanism
- Successful proof of concept for a **scalable, efficient, and practical egg extermination method**
- Future iterations will involve refinement of **spiral geometry**, **reduction of scale** for most compact design, and improved efficiency through **multiple built-in crushing mechanisms**

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References:

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